

BioAro Drugs AI Model Handoff v2

Audience: AI, backend, product, frontend, data, safety, regulatory, and DevOps teams **Status:** Production planning and implementation handoff **Canonical source:** This Markdown file **Last reviewed:** 2026-08-18 **Supersedes for implementation planning:** bioaro-ai-model-handoff.md

1. Executive Summary

BioAro Drugs AI is a 360-degree everyday-health assistant combining advanced BioAro knowledge with guidance shaped by clinical experience. It helps users understand approved product information, interpret non-clinical goals, build a structured routine, and return to refine it over time.

The product model is **Ask -> Build -> Evolve**:

- **Ask:** approved product, ingredient, usage, availability, and educational information.
- **Build:** a starting protocol assembled from stated goals and structured answers.
- **Evolve:** a persistent member workspace where goals, conversations, and routines can be revisited.

The model may interpret language and produce structured candidates. It must never own product facts, prices, inventory, medical decisions, partner eligibility, or membership entitlement. Backend policy and approved data remain authoritative.

Release boundary

The current frontend has deterministic keyword matching, approved local retrieval, deterministic protocol rules, browser-only conversation history, browser-only attachment previews, and local demo membership. Production launch requires server-side retrieval, safety policy, structured-output validation, authenticated persistence, verified entitlement, and approved regulatory content.

2. Product Positioning And Boundaries

Public AI

Public AI is for exploration: product questions, ingredient information, usage, regional availability, basic educational answers, and starting a goal conversation.

Membership

Membership provides continuity: the advanced workspace, guided protocol building, saved conversations, a daily routine view, and ongoing refinement. It is not medical advice and must not be framed as clinically superior intelligence.

Explicit exclusions

The system must not diagnose, prescribe, interpret prescriptions, recommend drug interactions, interpret biomarkers or genetics, make clinical decisions, guarantee outcomes, or replace a doctor, pharmacist, or qualified healthcare professional.

3. Current Implementation Versus Production Target

Area	Implemented in the current app	Production requirement
Q&A;	Approved local retrieval and deterministic fallback	Server retrieval with citations and policy validation
Intent	Keyword goal matching and frontend result contracts	Validated classifier/model over the complete intent catalogue
Protocol	Deterministic buildProtocol rules	Model-assisted clarification, deterministic product validation
Safety	Client classifier for approved copy	Server-first safety precedence and refusal pipeline
Catalogue	Local editorial fallback plus Shopify mapping	Versioned live catalogue with price, stock, market, and claim controls
Membership	Demo snapshot or unavailable backend	Authenticated backend entitlement snapshot
History	Browser localStorage; resumable local records	Account-bound encrypted persistence and deletion controls
Attachments	Browser object URLs; no upload, OCR, or interpretation	Explicit storage boundary; no prescription interpretation in V1
Partners	Registry contract; BioSports, Biogevity, and BioAro Labs draft	Approved registry plus backend eligibility policy

4. User Journeys And Entry Points

```
flowchart LR
    H[Homepage] --> Q[Public Q&A]
    H --> G[Goal input or chip]
    G --> I[Intent validation]
    I --> S[Protocol Studio]
    S --> A[Advanced member workspace]
    Q --> R[Approved answer, referral, or clarification]
    A --> P[Protocol and routine refinement]
    P --> N[Optional partner policy check]
    AC[Account or membership page] --> A
```

Supported entry points are the homepage AI field, homepage goal chips, floating AI control, header AI navigation, account dashboard, membership page, previous-chat resume, and direct `/:market/ai` visits. A public question does not automatically force a membership purchase. A protocol-building handoff may open the advanced route, where non-members see the membership gate.

5. Complete Intent Catalogue

The classifier must use the following controlled IDs. This is the complete **50-intent vocabulary** from the AI team's routing contract. Product, route, goal, and safety are policy families, not extra IDs; an ID may belong to more than one family and is listed once in the classifier registry. Every intent has an action, response mode, safety check, and fallback. Unknown or low-confidence input must not be forced into a known class.

Product and formula intents

product_information, ingredient_information, supplement_facts, dosage_directions, timing_usage, product_compatibility, dietary_allergen, warnings_precautions, expected_results_timeline, compare_products, product_format_convenience, pricing_availability, stock_availability, product_reviews_intent, subscription_reorder_intent, certificate_batch_intent, testing_quality_intent, unsupported_product_claim_intent.

Goal and outcome intents

energy_vitality, longevity_healthy_aging, focus_cognitive_performance, strength_muscle_performance, recovery_endurance, sleep_relaxation, heart_brain_cellular_wellness, womens_energy_vitality, antioxidant_skin_cellular_support.

Product-specific intents

The seven canonical product handles currently covered by the intent contract are:

Product intent family	Canonical handle
LONgevity+	longevity-plus
Glutara	glutara
Creagen Raw Power	creagen-raw-power
Creagen Pro Power	creagen-pro-power
Creagen Femme Energy	creagen-femme-energy
Creagen Brain Boost	creagen-brain-boost
CellOmega+	cellomega-plus

Product-specific intent IDs must resolve by handle, never by generated product name.

Route, membership, and safety intents

Route-linked intents include the 12 existing market-aware navigation destinations: shop/catalogue, quiz/protocol builder, science, journal, about, support, FAQ, disclaimer, account, membership, AI workspace, and partner/service navigation. Safety-linked intents include pregnancy/breastfeeding, medication interaction, diagnosed condition, urgent safety, unsupported claim, and professional-guidance requests. `unsupported_product_claim_intent` is intentionally shared by the product and safety families, not counted twice. Exact route names remain market-aware and must use the existing route helper.

6. Intent-To-Action Matrix

Intent family	Backend action	Response mode	Maximum UI output	Fallback
Product information	Retrieve approved product fields	answer	One answer plus source	no-match
Ingredient/supplement facts	Retrieve exact approved field and dose	answer	One answer plus source	Never guess; no-match
Usage/timing	Retrieve label directions and approved usage copy	answer	One answer plus disclaimer	Professional guidance if sensitive
Compatibility/allergen/warnings	Policy check, then retrieve warnings	answer or sensitive	No product recommendation on blocked state	Referral
Price/stock/availability	Read live Shopify/market catalogue	answer	One product or max two comparisons	unavailable
Compare products	Retrieve two approved handles	answer	Maximum two products	Clarification
Goal/outcome	Map goal to ranked approved products	needs-quiz, protocol, or answer	Maximum two product cards	Clarification/no-match
Product-specific goal	Product intent wins after safety check	answer or protocol	Requested product plus one alternative at most	Product clarification
Route/navigation	Return market-aware route	route	One destination CTA	fallback_unknown_intent
Membership/commerce	Read membership/offer state	route or answer	One primary CTA	Support
Safety/clinical	Stop product and partner pipeline	sensitive or referral	Safety copy and handoff only	Emergency-specific copy
Partner service	Run registry policy endpoint	answer with partner field	One or two subordinate cards	Suppress

routeSummary alone is not an answer. Informational intents must use the additive answer mode and include grounded sources.

7. Goal-To-Product Recommendation Matrix

This table is the initial deterministic fallback and must be reviewed by Product and Regulatory before production. The model cannot invent a mapping.

Goal	Primary approved starting handle	Secondary candidate	Rule
energy_vitality	cellomega-plus	creagen-femme-energy only when explicitly relevant	Prefer foundational support; no medical claim
longevity_healthy_aging	longevity-plus	cellomega-plus	Healthy-ageing language only
focus_cognitive_performance	creagen-brain-boost	cellomega-plus	Use focus/product facts, not cognitive diagnosis
strength_muscle_performance	creagen-pro-power	creagen-raw-power	Training context required for protocol refinement
recovery_endurance	creagen-pro-power	creagen-raw-power	Do not imply injury treatment
sleep_relaxation	No approved sleep formula	magba lance only if approved for the market	Return honest no-match when unavailable
heart_brain_cellular_wellness	cellomega-plus	longevity-plus	Use approved cellular/omega facts only
womens_energy_vitality	creagen-femme-energy	cellomega-plus	Explicit goal, not inferred sex or diagnosis
antioxidant_skin_cellular_support	glutara	cellomega-plus	Claims require market approval

Tie-break rules: safety status first; explicit product request second; explicit goal order third; market availability fourth; approved primary ranking fifth. Show no more than two products. A product unavailable in the selected market is removed, not replaced by an unrelated product.

8. Input Contracts And Validation

```

interface InterpretRequest {
  message: string;
  selectedGoals: GoalId[];
  market: MarketCode;
  sessionContext?: AllowlistedSessionContext;
}

interface AskRequest {
  query: string;
  market: MarketCode;
  selectedGoals?: GoalId[];
  conversationId?: string | null;
}

interface ProtocolRequest {
  market: MarketCode;
  goals: GoalId[];
  answers: PartialAnswers;
  clinicalFlag: boolean;
  existingSupplements?: SupplementsAnswer;
  sessionId?: string;
}

interface PartnerContext {
  market: MarketCode;
  goals: GoalId[];
  nonClinicalSignals: string[];
  explicitServiceInterest?: string;
  previousPartnerIds?: string[];
  safetyStatus: "allowed" | "referral" | "blocked";
}

```

Validation rules: market must be uk, us, ca, or ae; messages must be trimmed and capped at 200 characters for intent input and the agreed Q&A; limit for production; goals must be allowlisted and deduplicated; session context must be allowlisted; raw clinical, medication, prescription, or attachment content must not enter partner context; prompt-injection text never changes policy.

9. Adaptive Questions And Session Lifecycle

```

interface ProtocolQuestion {
  id: string;
  field: AnswerField;
  stage: "about" | "clinical" | "precision";
  prompt: string;
  options: QuestionOption[];
  appliesTo?: GoalId[];
}

```

Ask one question at a time and stop after three clarification questions. Ask only when the answer can change product selection, removal, timing, rationale, or safety copy. Clinical questions may trigger referral; they must never act as recommendation intelligence. If a user changes goals, preserve the new explicit order, recalculate the protocol, and remove products no longer supported. A session expires after 30 days of inactivity; authenticated members may resume from backend storage, while local demo sessions remain browser-only.

10. Public Q&A; And Intent APIs

POST /api/bioaro-ai/interpret

```
interface InterpretResponse {
  interpretedGoals: GoalId[];
  confidence: number;
  summary: string;
  suggestedNextQuestion?: string;
  builderContext?: { primaryGoals: GoalId[] };
  requestId: string;
  sessionId?: string;
}
```

Confidence must be numeric and bounded from 0 to 1. Recommended initial behavior: confidence below 0.65 returns clarification, not a product.

POST /api/bioaro-ai/ask

```
interface AskAnswer {
  question: string;
  answer: string;
  sources: AskSource[];
}

interface AskSource {
  kind: "product" | "faq" | "journal";
  label: string;
  href: string;
  sourceVersion?: string;
}
```

The endpoint returns answer, needs-quiz, sensitive, no-match, or error. Every answer must be traceable to approved retrieval. Prices, inventory, doses, and health claims are never generated by the model.

POST /api/bioaro-ai/protocol

```
interface ProtocolItem {
  handle: string;
  slot: "Morning" | "Around training" | "Evening";
  reason: string;
}

interface Protocol {
  items: ProtocolItem[];
  notes: string[];
}
```

The backend validates handles, market visibility, source-backed reasons, overlap removal, and safety state before returning the protocol.

11. Combined Response Envelope

```
interface BioAroAiResponse {
  requestId: string;
  sessionId?: string;
  market: MarketCode;
  kind: "answer" | "protocol" | "clarification" | "route" | "sensitive" | "no-match" | "error";
  summary?: string;
  confidence?: number;
  answers?: AskAnswer[];
  protocol?: Protocol;
  nextQuestion?: ProtocolQuestion;
  partnerRecommendations?: PartnerRecommendation[];
  disclaimers: string[];
  safety: {
    status: "allowed" | "referral" | "blocked";
    clinicalReviewRequired: boolean;
    reasonCode?: string;
  };
  provenance: {
    model: string;
    retrievalSources: string[];
    policyVersion: string;
  };
}
```

`requestId` supports support tracing; `sessionId` supports multi-turn continuation; `market` prevents cross-market leakage; `kind` selects the UI; `confidence` supports thresholds and evaluation; `disclaimers` makes safety copy explicit; `provenance` supports auditability; and `partnerRecommendations` remains optional and policy-controlled.

12. Mandatory Safety Precedence

```
flowchart TD
  IN[Input] --> E{Emergency or crisis?}
  E -->|Yes| CR[Crisis or emergency response]
  E -->|No| C{Clinical, pregnancy, medication, prescription, or condition context?}
  C -->|Yes| REF[Referral or blocked response]
  C -->|No| U{Unsupported claim or prompt injection?}
  U -->|Yes| SAFE[Safe refusal or approved disclaimer]
  U -->|No| INT[Resolve intent and goals]
  INT --> RET[Approved retrieval and protocol policy]
  RET --> PART[Partner policy check]
  PART --> OUT[Schema, claims, source, and disclosure validation]
```

Safety is evaluated before intent resolution and can hard-stop product, protocol, price, and partner outputs. Mixed prompts such as "I take medication and need energy" must enter the clinical branch. Emergency and self-harm handling require separate approved copy and resource paths. The interaction matrix, crisis copy, claims lexicon, age policy, and jurisdiction-specific disclaimer text are launch blockers requiring Regulatory/Clinical sign-off.

Required controlled safety data

- Clinically reviewed drug-supplement interaction matrix; unknown combinations fail closed.
- Market-specific banned-claims lexicon and outgoing claims validator.
- Emergency and self-harm response copy with regional resources.
- Pregnancy, breastfeeding, diagnosed-condition, and medication policy.
- Minor/under-18 policy for product and protocol flows.
- Named accountable approver for each safety and claims release.

Engineering must not author the interaction matrix or clinical policy without clinical ownership.

13. Retrieval And Shopify Data Model

Shopify and approved editorial data are the source of truth for product title, handle, ingredients, dose, supplement facts, warnings, quality claims, FAQs, market availability, price, and stock. The local TypeScript catalogue is a fallback for preview only.

Each retrieval document must carry `documentId`, `sourceType`, `productHandle` or `partnerId`, `market`, `content`, `sourceUrl`, `sourceVersion`, `allowedUses`, `blockedUses`, and `visibility`. Product facts and partner content must be indexed separately. The model returns a handle or answer candidate; a backend resolver retrieves and validates the actual content.

The AI may not control Shopify price, checkout, publication, inventory, or membership entitlement.

14. Partner-Service Promotion

```
interface PartnerService {
  id: string;
  name: string;
  description: string;
  category: "sports-performance" | "healthy-ageing" | "testing-and-insights" | "professional-support" | "other";
  approvedMarkets: MarketCode[];
  approvedGoals: GoalId[];
  approvedAnswerSignals: string[];
  excludedSignals: string[];
  url: string;
  ctaLabel: string;
  disclosure: string;
  status: "active" | "draft" | "paused";
}
```

BioSports, Biogevity, and BioAro Labs remain draft and are not shown until their descriptions, URLs, markets, approved signals, exclusions, and disclosure are approved. Partner matching may use explicit goals, non-clinical lifestyle signals, training/performance interests, healthy-ageing interests, testing/evidence interests, and explicit service requests. It must never use diagnoses, symptoms, medications, pregnancy, clinical flags, biomarkers, genetics, inferred risk, or protected health information.

```
interface PartnerRecommendation {
  partnerId: string;
  name: string;
  reason: string;
  ctaLabel: string;
  href: string;
  disclosure: string;
  confidence: number;
  eligibility: { market: boolean; goal: boolean; policy: boolean };
}
```

POST /api/bioaro-ai/partner-recommendations returns at most two recommendations. The endpoint enforces status, market, goal, safety, disclosure, and count limits; logs policy decisions without raw sensitive answers; and suppresses all recommendations when safety is referral or blocked.

15. Membership And Entitlement

```
Shopify purchase -> success page -> backend membership refresh
-> verified snapshot -> hasAdvancedAiAccess() -> /:market/ai
```

Advanced access requires membership.status === "active" and tierId in essential or plus. A checkout redirect or query parameter never grants access. Proposed authenticated APIs are GET /api/member/snapshot, conversation and protocol CRUD endpoints, and attachment endpoints. All records require account ownership and market isolation.

16. Conversation And Attachment Handling

```
interface Conversation {
  id: string;
  market: MarketCode;
  title: string;
  startedAt: string;
  updatedAt: string;
  turns: Turn[];
  session?: ProtocolSession;
}
```

The current browser history can resume local text and saved protocol state; images are never persisted. Production history must be authenticated backend data with account isolation, market isolation, deletion behavior, retention policy, and audit logging. Attachments require an explicit future storage boundary. Label and prescription intent may be selected, but V1 performs no upload, OCR, image interpretation, prescription interpretation, or medication advice.

17. Hosting, Security, And Performance Decisions

Before model integration, Engineering and DevOps must record the chosen provider/model, hosting region, auth method, CORS policy, secret storage, gateway, per-IP and per-account rate limits, abuse controls, timeout, retry policy, latency budget, caching policy, and cost budget. The public Q&A endpoint must not expose provider credentials and must not be an unrestricted browser-to-LLM proxy.

Recommended initial targets: p95 response under 2.5 seconds for public answers, p95 under 4 seconds for protocol refinement, bounded retries, cache approved repeated Q&A; by normalized query plus market and source version, and fail closed when policy or catalogue services are unavailable.

18. Error And Fallback Matrix

Failure	HTTP/status behavior	User result	Blocked?
Empty or oversized input	400	Ask for a shorter question	Yes, no model call
Unknown or low-confidence goal	200 clarification	Ask one focused question	No product
Sensitive/clinical request	200 referral	Approved professional-guidance copy	Product/partner blocked
Emergency/self-harm	200 urgent safety	Regional emergency/crisis guidance	All recommendations blocked
No retrieval match	200 no-match	Explain limits and offer support/search	No invention
Invalid model JSON	502 internally	Deterministic safe fallback	Output blocked
Unknown product handle	422 internally	No product card	Yes
Unsupported market	400	Market unavailable message	Partner/price blocked
Missing claim/disclosure/source	422 internally	Suppress offending output	Yes
Shopify/catalogue unavailable	503	Availability temporarily unavailable	No guessed price/stock
Membership unavailable	503	Retry/support gate	No unlock
Attachment rejected	415	Explain file limitation	No persistence
Rate limit/timeout	429/504	Retry later	Request blocked
Conversation save failure	503 partial	Continue current session, show persistence warning	No data loss claim

19. Privacy And Analytics

Track only non-sensitive events: opened, message sent, intent resolved, answer shown, product clicked, add-to-cart, purchase, safety referral, fallback, handoff, partner recommendation, suppression, partner CTA click, protocol completion, resume success, latency, and error. Scrub message text, medications, prescriptions, conditions, clinical flags, and attachment content before analytics. Never place health answers in URLs. Define retention, deletion, consent, access, and data-processing rules for each launch market.

20. Evaluation And Launch Gates

Required evaluation corpus categories include single-goal, multi-goal, ambiguous, product, ingredient/dose, availability, comparison, safety, crisis, prompt injection, overlap, sleep limitation, partner matching/suppression, unsupported markets, invalid model output, checkout verification, resume, attachment non-persistence, and cross-account isolation.

Initial launch gates, to be confirmed by owners:

Metric	Initial gate
Safety recall	1.00 on approved safety corpus before launch
Claims violation rate	0 on release-candidate corpus
Product grounding	100% of displayed handles and facts traceable to approved sources
Intent performance	Precision/recall threshold agreed per intent; no forced low-confidence class
Partner suppression	100% suppression for clinical, unsupported-market, paused, or missing-disclosure cases
Access control	100% rejection for unauthenticated/ineligible advanced workspace requests
Isolation	0 cross-account or cross-market records in test corpus
Latency	p95 within agreed public and protocol budgets

21. Ownership And Approval Register

Decision	Required owner	Status
Goal-to-product ranking	Product + Regulatory	Open
Product factual content	Product/Data	Shopify and approved content required
Interaction matrix	Clinical/Pharmacist	Blocking; not engineering-owned
Crisis and sensitive copy	Clinical/Regulatory	Blocking
Claims lexicon and disclaimers	Regulatory	Blocking per market
Age/minor policy	Product + Regulatory	Open
Partner metadata and disclosure	Partnerships + Legal	Draft; partners disabled
Model/hosting/latency/cost	AI + DevOps	Open
Auth, persistence, entitlement	Backend + Security	Required before production
Analytics/privacy retention	Data + Legal	Required before launch

22. Implementation Checklist

- Confirm launch markets and regulatory owners.
- Approve the 50-intent catalogue and action matrix.
- Approve goal-to-product rankings and tie-break rules.
- Provide the clinically reviewed interaction matrix.
- Approve crisis, pregnancy, medication, age, and claims policies.
- Define market-specific disclaimers and banned-claims validators.
- Confirm Shopify/catalogue source and source-versioning strategy.
- Confirm backend hosting, auth, rate limits, caching, and observability.
- Implement /interpret, /ask, and protocol services.
- Implement safety and structured-output validation before UI rendering.
- Implement partner registry and policy endpoint with all initial partners disabled.
- Implement authenticated membership, conversation, protocol, and attachment boundaries.
- Add evaluation corpus, red-team tests, and launch dashboards.
- Launch behind a controlled feature flag.

23. Open Decisions Requiring Sign-Off

The following must not be silently decided by the implementation team: whether Q&A; answers are retrieval text or grounded generation; exact model/provider; all goal ranking and bundle behavior; the maximum comparison/product-card count; the missing reviews and reorder routes; pregnancy/medication response policy; interaction matrix owner; crisis resources; age policy; market/legal scope; backend deployment; identity binding; retention; and partner commercial terms.

24. Interface Inventory

Existing frontend contracts

InterpretRequest, InterpretResponse, InterpretResult, AskResult, AskAnswer, ProtocolQuestion, ProtocolAnswers, ProtocolItem, Protocol, Conversation, Turn, MemberRequest, MemberSnapshot, and MembershipOffer.

Proposed backend contracts

AskRequest, ProtocolRequest, PartnerService, PartnerContext, PartnerRecommendation, BioAroAiResponse, AttachmentRequest, and AttachmentResponse.

Sensitive/private fields

Clinical flags, prescription references, medication context, account identity, member conversations, attachment metadata, and raw answers. These must not be used for uncontrolled partner promotion or sent to analytics by default.

Appendix A - Reference Flows

Public Q&A; sequence

```
sequenceDiagram
    participant U as User
    participant W as Web app
    participant API as AI API
    participant K as Approved knowledge
    participant P as Safety policy
    U->>W: Ask question
    W->>API: query + market + context
    API->>P: classify safety and injection
    P-->>API: allowed or referral
    API->>K: retrieve approved sources
    K-->>API: versioned facts
    API-->>W: validated answer + sources + safety
    W-->>U: Render answer or safe referral
```

Partner decision flow

```
flowchart LR
    C[Non-clinical context] --> R[Partner registry]
    R --> M{Market and goal eligible?}
    M -->|No| S[Suppress]
    M -->|Yes| X{Safety allowed and disclosure present?}
    X -->|No| S
    X -->|Yes| O[One or two subordinate partner cards]
```

Appendix B - Visual References

Use the existing repository imagery only as interface context, not as model evidence:

[Image reference: Product context]

[Image reference: Protocol context]

[Image reference: Evidence context]

The review package should include current screenshots of `/:market/ai`, Previous chats, membership, public Q&A;, and partner suppression states. Partner cards must be visually subordinate to BioAro product answers and clearly labelled as partner services.

Final Safety Position

BioAro Drugs AI may help users understand approved information, structure non-clinical goals, assemble a grounded routine, return to previous work, and discover approved partner services. It cannot determine what is medically right for a person. Production launch requires every answer, product, price, source, partner, and entitlement to be traceable to an approved system of record, with safety precedence and clinical/regulatory ownership in place.